

SwagRestartScheduler

Plugin Specification Document

Developer: swag617

Platform: Paper/Spigot (Minecraft 1.20+)

Language: Java

Type: Server Utility Plugin

Project Overview

SwagRestartScheduler is a restart scheduling plugin for Paper/Spigot servers that provides advanced scheduling capabilities with timezone support, pre-restart automation, and Discord integration. The plugin prioritizes ease of configuration through an in-game GUI while maintaining powerful scheduling features for complex server environments.

Core Features

1. Advanced Scheduling System

Multiple Schedule Support:

- Define multiple named schedules (e.g., "weekday", "weekend", "event")
- Each schedule can have multiple restart times per day
- Support for different schedules based on day of week
- Ability to enable/disable schedules without deleting them

Timezone Management:

- Full timezone support using Java's `ZoneId`
- Configure schedules in any timezone (e.g., EST, PST, UTC, etc.)
- Display times to players in their configured timezone or server timezone
- Timezone conversion for multi-region communities

Schedule Configuration Format (YAML):

```
schedules: weekday: enabled: true timezone: "America/New_York" days: [MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY] times: - "03:00" # 3 AM EST - "15:00" # 3 PM EST weekend: enabled: true timezone: "America/New_York" days: [SATURDAY, SUNDAY] times: - "05:00" # 5 AM EST
```

2. Warning System

Countdown Broadcasts:

- Configurable warning intervals (default: 30min, 15min, 5min, 1min, 30sec, 10sec)
- Full MiniMessage formatting support for warnings
- Different message formats for different time intervals
- Action bar countdown display during final minute
- Title/subtitle support for critical warnings (< 1 minute)

Warning Configuration:

```
warnings: enabled: true intervals: - seconds: 1800 # 30 minutes message: "<gold>■</gold> <yellow>Server restart in <white>30 minutes</white>!" title: null subtitle: null - seconds: 60 # 1 minute message: "<red>■</red> <yellow>Server restart in <white>1 minute</white>! Save your progress!" title: "<red><bold>RESTARTING" subtitle: "<yellow>1 minute remaining" - seconds: 10 message: "<dark_red>Server restarting in 10 seconds!" title: "<dark_red><bold>RESTARTING" subtitle: "<red>10 seconds!" sound: "ENTITY_ENDER_DRAGON_GROWL"
```

3. Grace Period & Player Protection

Smart Delay System:

- Delay restart if players are in combat (requires CombatLogX or similar)
- Delay restart if players are in specific worlds/regions (WorldGuard integration)
- Maximum grace period limit (default: 15 minutes)
- Configurable conditions for grace period triggers

Grace Configuration:

```
grace_period: enabled: true max_delay_minutes: 15 conditions: combat: true # Delay if any
player is in combat worlds: - "world_boss" - "dungeon_*" # Wildcard support regions: #
Requires WorldGuard - "spawn_protection" - "pvp_arena" check_interval_seconds: 5 message:
"<yellow>Restart delayed - players in protected area"
```

4. Pre-Restart Automation

Command Execution:

- Execute commands before restart with configurable delays
- Support for console commands and plugin-specific commands
- Placeholder support via PlaceholderAPI
- Sequence multiple commands with delays

Automation Configuration:

```
pre_restart: enabled: true commands: - delay: 300 # 5 minutes before restart command:
"broadcast &c&lServer restarting in 5 minutes!" executor: "console" - delay: 60 command:
"backup start" executor: "console" - delay: 30 command: "staffmode enable-all" # Example:
Enable staff mode for all staff executor: "console" - delay: 5 command: "save-all" executor:
"console"
```

Plugin Integration Hooks:

- Direct integration with common backup plugins (CoreProtect, WorldEdit, etc.)
- Optional hooks for staff mode plugins
- BungeeCord/Velocity message support (notify other servers)

5. Discord Integration

DiscordUtils Integration:

- Send restart notifications to Discord channels
- Configurable embed formatting
- Include restart reason, time until restart, and player count
- Post-restart "server online" confirmation

Discord Configuration:

```
discord: enabled: true webhook_id: "your_webhook_id" # DiscordUtils webhook notifications:
scheduled_restart: enabled: true embed: title: "■ Scheduled Restart" color: "ORANGE"
description: "Server restarting in {time}" fields: - name: "Reason" value: "{reason}" - name:
"Players Online" value: "{player_count}" manual_restart: enabled: true embed: title: "■■
Manual Restart" color: "RED" server_online: enabled: true embed: title: "■ Server Online"
color: "GREEN" description: "Server has restarted successfully"
```

6. Manual Restart Commands

Command Suite:

```
/restart now [reason] - Restart immediately /restart in <time> [reason] - Schedule one-time
restart (e.g., "30m", "1h", "45s") /restart cancel - Cancel pending restart /restart schedule
```

```
<name> - View or trigger a named schedule /restart schedules - List all configured schedules  
/restart status - Show next scheduled restart /restart reload - Reload configuration
```

Permission Nodes:

```
swagrestart.command.now swagrestart.command.schedule swagrestart.command.cancel  
swagrestart.command.reload swagrestart.bypass.grace # Bypass grace period delays
```

7. In-Game GUI Configuration

GUI Features:

- View all configured schedules
- Enable/disable schedules with click
- Create new schedules (name, timezone, days, times)
- Edit existing schedules
- Test warning messages
- View next restart time for each schedule

GUI Structure:

```
Main Menu: - [Clock] View Schedules - [Calendar] Create Schedule - [Wrench] Settings - [Book]  
Restart Logs - [Bell] Test Warnings Schedule Editor: - [Green Wool] Enable/Disable toggle -  
[Clock] Edit times (click to add/remove) - [Globe] Change timezone (searchable) - [Calendar]  
Select active days - [Chest] Pre-restart commands - [Discord] Discord settings
```

8. Logging & Analytics

Restart Tracking:

- Log all restarts with timestamp, reason, and initiator
- Track restart duration and post-restart verification
- Player count at restart time
- Export logs to CSV for analysis

Log Format:

```
logs: - timestamp: "2025-03-24T15:00:00Z" type: "scheduled" reason: "Daily restart" schedule:  
"weekday" initiator: "CONSOLE" players_online: 45 grace_period_used: true  
grace_duration_seconds: 120 restart_duration_seconds: 35
```

Optional Features (Nice-to-Have)

Player Vote Delay System

```
vote_delay: enabled: true threshold_percent: 75 # % of players must vote yes delay_minutes: 15  
max_uses_per_restart: 1 command: "/restart vote"
```

Performance-Based Triggers

```
performance_triggers: enabled: false tps_threshold: 15.0 duration_minutes: 5 # TPS below
threshold for this long action: "schedule" # or "immediate" reason: "Performance degradation
detected"
```

Multi-Schedule Priority

```
# If multiple schedules overlap, use priority schedules: high_priority: priority: 1 # ...
low_priority: priority: 10 # ...
```

Technical Requirements

Dependencies

Required:

- Paper API 1.20+
- Java 17+

Optional (Soft Dependencies):

- DiscordUtils (for Discord integration)
- PlaceholderAPI (for placeholder support)
- WorldGuard (for region-based grace periods)
- CombatLogX (for combat detection)
- Vault (for permission checks)

Plugin Architecture

Main Components:

1. **ScheduleManager** - Handles schedule loading, timezone conversion, next restart calculation
2. **RestartTask** - Bukkit runnable that manages restart countdown and execution
3. **WarningManager** - Broadcasts warnings at configured intervals
4. **GracePeriodHandler** - Checks grace period conditions and delays restart if needed
5. **CommandExecutor** - Executes pre-restart commands in sequence
6. **DiscordNotifier** - Sends Discord webhooks via DiscordUtils
7. **GUIManager** - Handles all in-game GUI interactions
8. **ConfigManager** - Manages YAML configuration and reload

Key Classes:

```
public class RestartSchedule { private String name; private boolean enabled; private ZoneId
timezone; private Set<DayOfWeek> activeDays; private List<LocalTime> restartTimes; private int
priority; // Methods: getNextRestart(), isActive(), etc. } public class RestartContext {
private RestartSchedule schedule; private String reason; private CommandSender initiator;
private LocalDateTime scheduledTime; private boolean useGracePeriod; private
List<PreRestartCommand> commands; }
```

Configuration Files

plugin.yml:

```
name: SwagRestartScheduler version: 1.0.0 main: com.swag617.restartsched.SwagRestartScheduler
api-version: 1.20 softdepend: [DiscordUtils, PlaceholderAPI, WorldGuard, CombatLogX, Vault]
commands: restart: description: Restart management commands usage: /restart
<now|in|cancel|schedule|status|reload> permission: swagrestart.command
```

config.yml: (Structure outlined above in feature sections)

schedules.yml: (Separate file for schedule definitions)

messages.yml: (All player-facing messages with MiniMessage support)

Implementation Priorities

Phase 1 (MVP):

1. Core scheduling system with timezone support
2. Warning broadcasts with MiniMessage
3. Manual restart commands (/restart now, /restart in, /restart cancel)
4. Basic YAML configuration
5. Restart logging

Phase 2 (Polish):

1. In-game GUI for schedule management
2. Grace period system with WorldGuard integration
3. Pre-restart command execution
4. DiscordUtils integration
5. Permission system

Phase 3 (Advanced):

1. Player vote delay system
2. Performance-based triggers
3. Multi-schedule priority handling
4. CSV export for analytics
5. Web-based config editor (static HTML + localStorage)

Code Style & Standards

- Follow Paper API best practices
- Use Adventure API for all text components (MiniMessage)
- Async operations where appropriate (file I/O, Discord webhooks)
- Comprehensive JavaDocs for public APIs
- Unit tests for ScheduleManager and timezone calculations
- Config validation on load with helpful error messages
- No `/reload` corruption issues - proper cleanup in onDisable()

SpigotMC Listing Notes

Key Selling Points:

- ■ Full timezone support (no more mental math for multi-region servers)
- ■ MiniMessage formatting for beautiful warnings
- ■ Smart grace period - won't restart mid-boss fight
- ■ Discord integration via DiscordUtils
- ■ In-game GUI - no YAML editing required
- ■ Pre-restart automation (backups, staff mode, etc.)
- ■ Zero performance impact when not restarting

Differentiation from competitors:

- Most restart plugins assume server timezone only
- Few support advanced grace period conditions
- GUI configuration is rare in this category
- Pre-restart command sequences are usually manual

Notes for Implementation

- **Timezone handling:** Use ``ZoneId`` and ``ZonedDateTime`` throughout, never raw ``Date`` or ``Calendar``
- **Warning precision:** Use ``System.currentTimeMillis()`` for exact countdown, not Bukkit scheduler ticks
- **Grace period:** Poll conditions every 5 seconds during grace window, timeout after max delay
- **Discord rate limits:** Queue messages if multiple restarts scheduled close together
- **Config reload:** Validate new config before applying, rollback on error
- **Thread safety:** SchedulerManager should be thread-safe for async Discord operations

Example Usage Scenarios

Scenario 1: Multi-region community

- US players see restart times in EST
- EU players see same restarts in CET
- Single schedule definition in config

Scenario 2: Weekend event server

- Different restart times on weekends
- Longer grace period during events
- Discord notifications to event channel

Scenario 3: Performance maintenance

- Scheduled restart at 3 AM daily
- Pre-restart: backup, clear entity counts, save chunks
- Post-restart verification webhook to Discord

End of Specification

This spec is ready for handoff to Claude CLI. All core features, optional features, technical architecture, and implementation priorities are defined. Begin with Phase 1 (MVP) and test thoroughly before moving to Phase 2.